

Indian Electricity Datasets for the Proposed DM Project

Proposed project: Categorization of Residential Electricity Consumption Patterns Using Clustering and Classification Techniques.

The list below contains the Indian datasets discussed for the project, with direct links and a suitability assessment for our planned clustering + classification workflow.

1. Mathura & Bareilly Smart Meter Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/pythonafroz/electricity-smart-meter-data-from-india>

What it contains: Indian smart-meter data from Mathura and Bareilly. Multiple meters, timestamps and electricity-consumption measurements, with additional electrical variables.

Suitability: BEST PRACTICAL CHOICE

Why: Multiple Indian meters make consumer-level clustering possible. High-frequency readings allow us to construct daily/hourly consumption profiles. Manageable for a student project.

2. Prayas eMARC

Source: Prayas Energy / Harvard Dataverse

Link: <https://energy.prayas pune.org/our-work/article-and-blog/emarc-insights-from-smart-metering-data>

What it contains: Smart-meter data from 115 Indian households, with minute-wise electricity information and household/appliance survey information.

Suitability: BEST RESEARCH POTENTIAL

Why: Very rich dataset. Besides consumption patterns, household and appliance information can help explain why clusters differ. Main drawback is that the raw data may require more work to understand and process.

3. IIT Bombay Residential Electricity Dataset

Source: Figshare

Link: <https://figshare.com/s/a4aedbe2ae2aadc5618b>

What it contains: Residential electricity measurements from apartments on the IIT Bombay campus. The public version contains data for multiple apartments at hourly resolution.

Suitability: BEST BEGINNER OPTION

Why: Small and comparatively easy to handle. Multiple apartments are available, so clustering consumers is possible. Less data than the first two options.

4. Indian Residential Electricity Field Experiment

Source: ScienceDirect / open-access article

Link: <https://www.sciencedirect.com/science/article/pii/S2352340917306807>

What it contains: Detailed electricity data from 19 Indian apartments, with apartment characteristics and weather information.

Suitability: GOOD BACKUP

Why: Useful for combining consumption with household and weather information. However, only 19 apartments makes the clustering study relatively small.

5. RESIDE-AC Dataset

Source: Figshare

Link: https://figshare.com/articles/dataset/dataset-Residential_Electricity_current_and_appliance_dataset_for_AC-event_detection_from_Indian_Dwellings_zip/16869439

What it contains: High-frequency residential electricity and AC-event data from Indian dwellings, with temperature, humidity and household information.

Suitability: GOOD FOR AN AC-SPECIFIC PROJECT

Why: Useful if the project is narrowed to cooling/air-conditioner consumption behavior. Not the best choice for broad residential consumer categorization because the number of homes and observation period are limited.

6. IRES 2020 - India Residential Energy Survey

Source: Kaggle / CEEW

Link: <https://www.kaggle.com/datasets/pythonafroz/electrical-consumption-data-in-indian-homes>

What it contains: Large Indian household survey covering nearly 15,000 households across many states, with appliance, household and energy-access information.

Suitability: GOOD FOR HOUSEHOLD SEGMENTATION

Why: Excellent number of households, but it is survey data rather than detailed smart-meter time-series data. Better for demographic/appliance-based segmentation than time-of-use consumption profiles.

7. Indian Household Electricity Bill Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/suraj520/indian-household-electricity-bill>

What it contains: Household, appliance and electricity-bill related variables.

Suitability: LOWER PRIORITY

Why: Easy to use, but it lacks the rich longitudinal smart-meter information needed for our main clustering idea.

Recommended order for our project

Rank	Dataset	Recommendation
1	Mathura & Bareilly	Best balance of multiple Indian meters, usable time-series data and manageable scope.
2	eMARC	Richest research opportunity, but more preprocessing/understanding may be needed.
3	IIT Bombay	Simplest option for beginners and a good prototype dataset.

Important: The Indian Mathura/Bareilly and eMARC datasets are alternatives to the private Karur/TANGEDCO dataset used in the 2024 Energy paper. We should not claim that they are the same dataset. They can instead be used to implement and extend the paper's general clustering/classification idea.